

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28 September 2025
Team ID	SWTID-2026-2056
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per Table-1 & Table-2.

This project implements Client Scripts and UI Policies on the ServiceNow Incident form to control field behaviour (mandatory, visible, read-only), validate data, auto-populate fields and provide real-time user feedback without requiring full page reloads.

Architecture Overview – Client Script & UI Policy (Incident)

Guidelines followed:

- Include all processes (Application Logic / Technology Blocks)
- Provide infrastructural demarcation (ServiceNow Cloud / Browser Client)
- Indicate external interfaces (GlideAjax, Script Includes, REST if used)
- Indicate Data Storage components / services (Incident table, sys_user, etc.)

High-Level Architecture Layers:

- Presentation Layer (Browser): ServiceNow Form UI – Desktop / Service Portal / Mobile
- Client-Side Logic Layer: Client Scripts (onLoad, onChange, onSubmit) + UI Policies + UI Policy Actions
- Platform API Layer: g_form, g_user, g_scratchpad, GlideAjax, GlideRecord (client)
- Server-Side Logic Layer: Script Includes, Business Rules, ACLs, Data Policies (for security)
- Data Layer: Incident (incident), User (sys_user), Configuration Item, Assignment Group tables
- Infrastructure: ServiceNow Cloud (SaaS) – multi-instance, multi-tenant platform

Note: Insert the architectural diagram (C4 Context / Container diagram or ServiceNow-specific form-flow diagram) in this section showing Browser → Client Script/UI Policy Engine → g_form API → (optional) GlideAjax → Script Include → Database.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How the user interacts with the Incident form – Desktop UI, Service Portal, Agent Workspace, Mobile	ServiceNow Form UI, AngularJS (Service Portal), Next Experience UI, HTML/CSS/JS
2	Application Logic – UI Policy	Declarative client-side rules that control field visibility, mandatory and read-only attributes based on conditions (e.g., State = Resolved) Supports Reverse if false and On load execution	ServiceNow UI Policy + UI Policy Actions (no-code / low-code)
3	Application Logic – Client Script	JavaScript executed in the browser on form events: onLoad, onChange, onSubmit, onCellEdit Used for complex logic, validation, colour changes, auto-populate, alerts	JavaScript (ES5/ES6 compatible), ServiceNow Client Script API (g_form, g_user)
4	Application Logic – Server Callback	Client Scripts call server-side Script Includes via GlideAjax for data lookup (Caller details, CI info) without full page reload	GlideAjax, Script Include (server-side JavaScript), JSON response
5	Platform APIs	Client-side APIs used by scripts and policies to manipulate the form and retrieve user/context data	g_form, g_user, g_scratchpad, g_navigation, GlideRecord (client-side)
6	Database / Data Store	Persistent storage of Incident records, user data, configuration items and related tables	ServiceNow MariaDB / MySQL-compatible RDBMS (Incident, sys_user, cmdb_ci, etc.)

7	Security Layer	Access control and data integrity enforced both client-side (UI Policy) and server-side (cannot be bypassed)	ACLs, Data Policies, Roles (itil, admin, ui_policy_admin), Business Rules
8	External Interface	Optional integration points (e.g., notification engines, external CMDB, or third-party APIs called from Script Includes)	REST Message, IntegrationHub, Outbound REST / SOAP (if required)
9	Infrastructure (Cloud)	Application runs entirely on ServiceNow SaaS platform No local server deployment required; multi-instance, multi-tenant architecture	ServiceNow Cloud (SaaS), multi-node application servers, load-balanced
10	Development / Admin Tools	Tools used to create, test and deploy Client Scripts and UI Policies	ServiceNow Studio, Form Designer, Application Navigator, Script Debugger, Browser DevTools

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	ServiceNow platform embeds open-source libraries for the client UI. Custom Client Scripts are written in standard JavaScript.	JavaScript, AngularJS (Service Portal), jQuery (legacy), Next Experience components
2	Security Implementations	Client-side UI Policies can be bypassed; critical rules enforced via ACLs and Data Policies Role-based access (itil, admin, ui_policy_admin) Instance-level encryption, SSO/SAML/OAuth support, audit logging	ACLs, Data Policies, Roles & Groups, SSO, Encryption at rest & in transit, OWASP-aligned platform controls
3	Scalable Architecture	ServiceNow is a multi-tenant, multi-node SaaS platform Client Scripts & UI Policies execute in the user's browser (scales with number of concurrent users) Server-side Script Includes are load-balanced across application nodes	ServiceNow multi-node architecture, horizontal scaling, browser-side execution
4	Availability	ServiceNow provides high availability with redundant application and database nodes Client-side logic continues to function as long as the browser session is active Platform SLA typically 99.8%+ uptime	ServiceNow HA clusters, load balancers, multi-AZ / multi-region options
5	Performance	UI Policies are lightweight (declarative, no heavy scripting) Client Scripts designed to be fast (<100 ms typical) Prefer GlideAjax (async) over synchronous GlideRecord Minimise number of onChange scripts on high-frequency fields	g_form API, GlideAjax (async), browser caching, ServiceNow CDN for static assets
6	Maintainability	Declarative UI Policies reduce code volume Clear naming conventions and short descriptions Scripts scoped to application; order controlled via Order field	ServiceNow Application Scope, Studio, Update Sets / Application Repository
7	Compatibility	Works on Desktop UI, Service Portal and Mobile (when UI Type = All). Compatible with current and future ServiceNow releases (Zurich and later).	ServiceNow UI Type setting, browser standards (Chrome, Edge, Firefox, Safari)